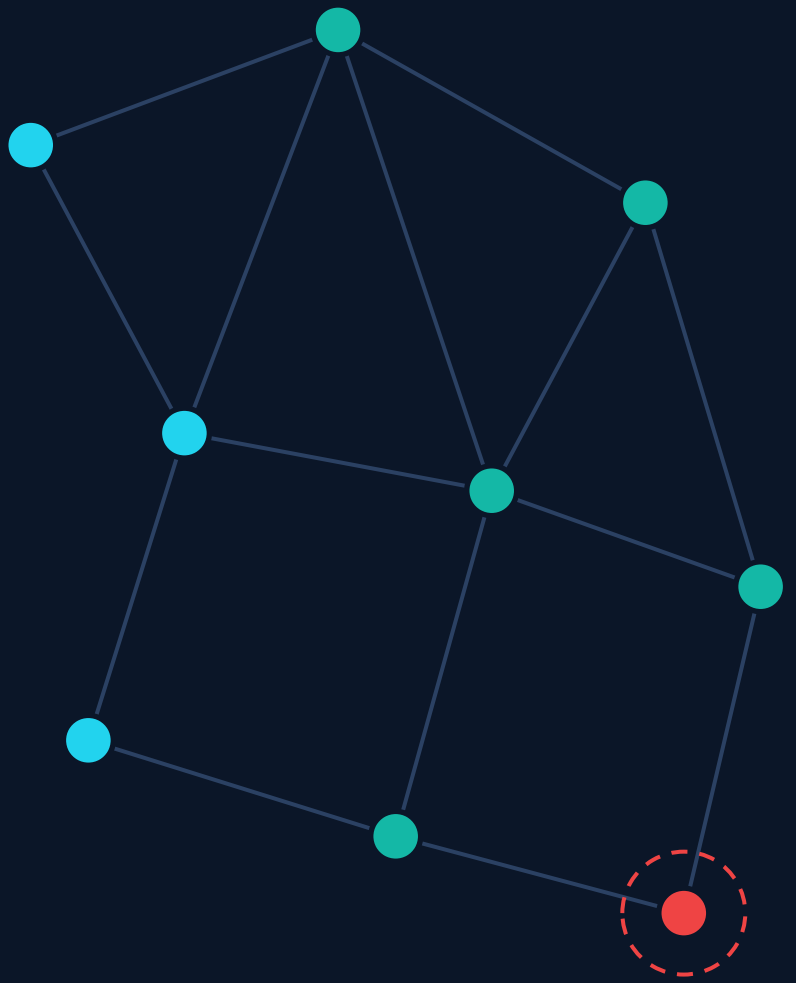
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COMPUTER NETWORKS PROJECT · ASSESSMENT-8

DNScope

DNS Monitoring and Spoofing Detection System

Observe. Compare. Investigate.

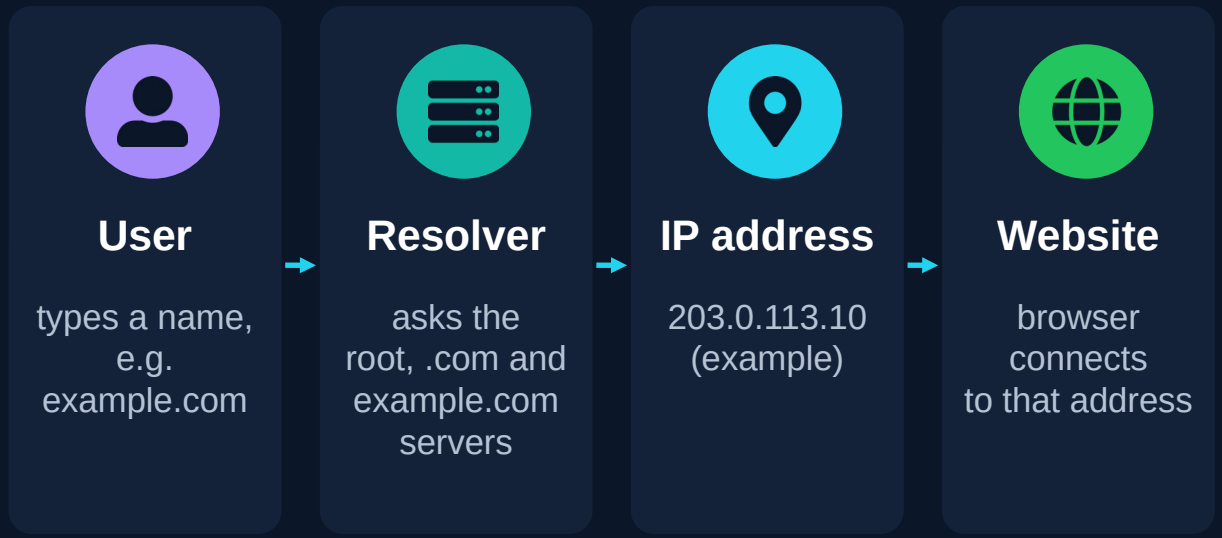


Anwita Padhi
Registration Number: 24BYB0193
Vellore Institute of Technology, Vellore · 2026–2027

What is DNS?

The Internet's phonebook

- **DNS = Domain Name System**
- Computers connect using IP addresses, people remember names
- DNS looks up a name and returns its IP address, like finding a number in a phonebook
- Almost every website visit, email and app connection starts with a DNS lookup



Under the hood: queries normally travel over UDP port 53 (TCP for large replies). Resolvers cache each answer for its TTL (time to live), so repeat lookups are fast.

Source: RFC 1034, RFC 1035

Problem statement

DNS answers can be manipulated

Spoofing or cache poisoning can redirect a name

Suspicious changes are hard to spot

Classic DNS over UDP has no built-in authentication

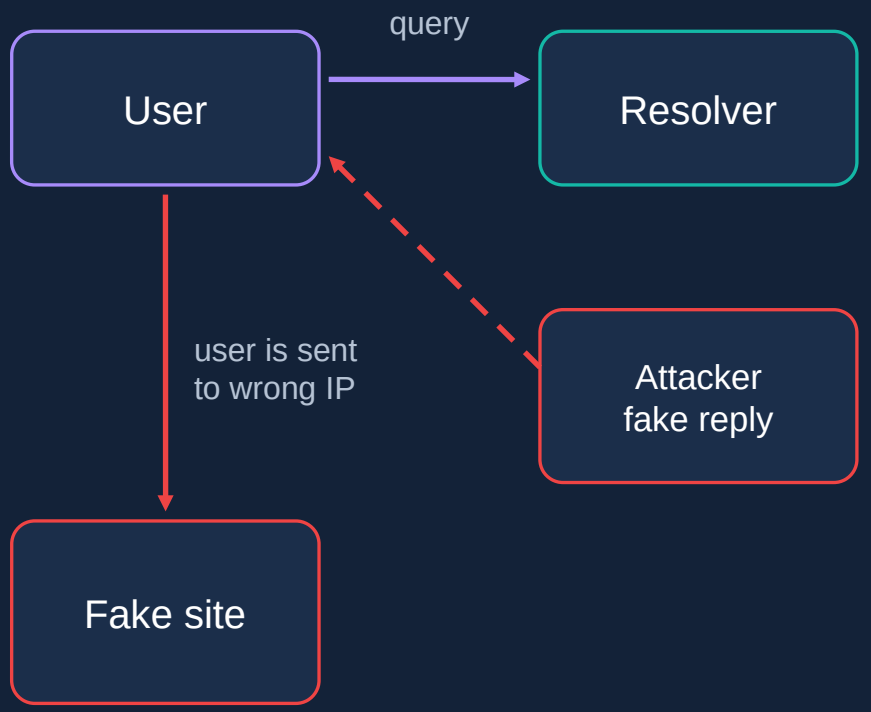
Lookup tools show one answer

dig / nslookup: one resolver, one moment, no history

Legitimate changes look similar

CDNs, load balancing, caching and migrations also change IPs

How a spoofed answer misleads



DNScope's aim: help users decide which DNS answers need investigation, without treating every change as an attack.

Proposed solution: how DNScope works

1



Accept domain

user enters a name; input is validated

2



Retrieve records

A, AAAA, CNAME, NS, MX, TXT via dnspython

3



Compare resolvers

system resolver vs 8.8.8.8, 1.1.1.1, 9.9.9.9

4



Check history

known IPs, network prefixes, usual TTL

5



Apply rules

transparent, configurable checks

6



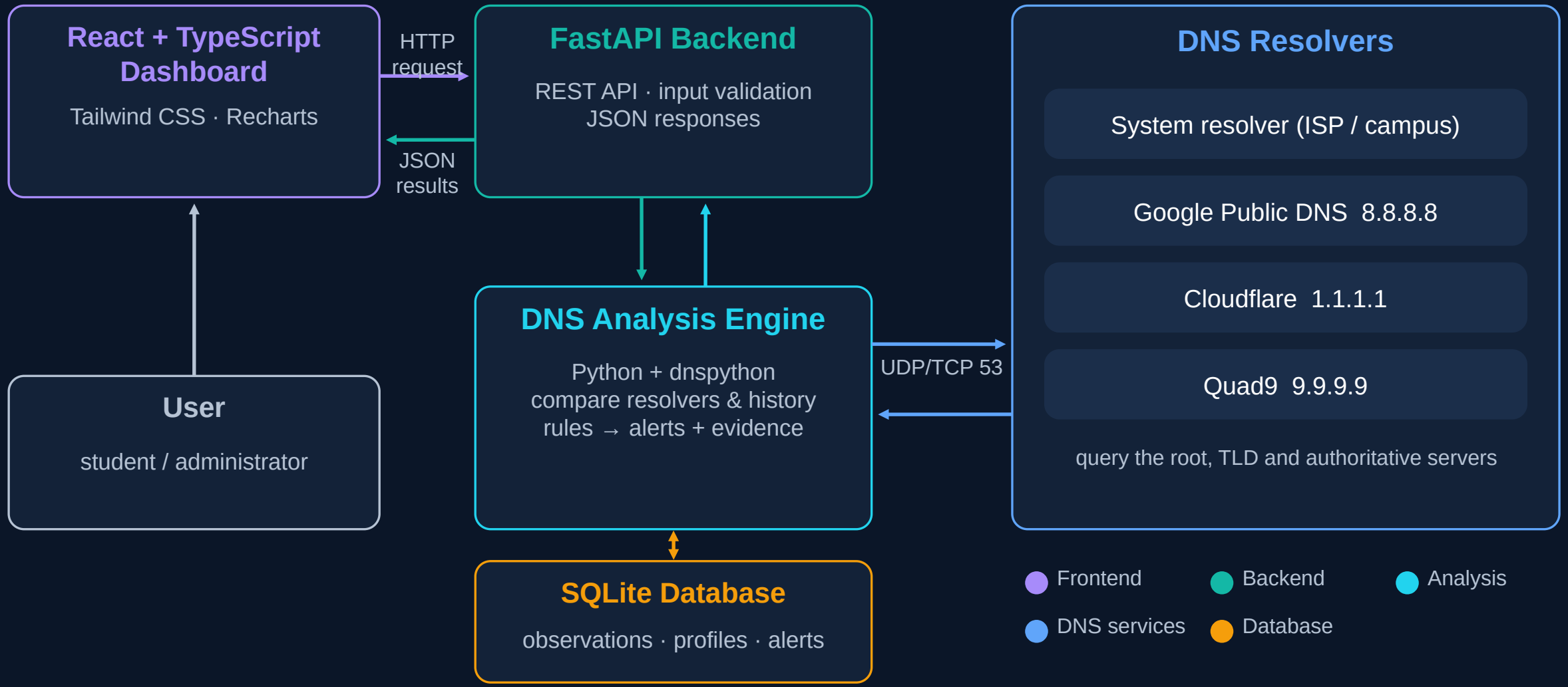
Show results

dashboard with evidence-based alerts

Example

Three resolvers return 198.51.100.20, but one returns 203.0.113.66 from an unrelated network with a much shorter TTL. DNScope flags it, shows the exact values, and lists harmless causes too (CDN routing, caching, a recent change).

System architecture



DNS detection methodology

Rule	Signal	What it checks	Why an alert may be raised
R1	**IP address change**	New IP not seen before and outside the domain's usual networks	Could be a redirect, or a legitimate migration
R2	**Resolver disagreement**	One resolver's answer shares nothing with the others	Could be poisoning, or CDN / geo-DNS routing
R3	**Unusual TTL**	TTL far shorter or longer than this domain's normal value	Attackers often use unusual TTLs; operators also change them
R4	**Repeated failures**	Timeouts, SERVFAIL or unexpected NXDOMAIN in several checks	Possible interference, or just an outage
R5	**DNSSEC failure**	Validating resolvers reject the data (when the zone is signed)	Data could not be authenticated
R6	**Private IP in answer**	Public name resolves to 10.x, 192.168.x or 127.x	Possible redirect to internal hosts



Signals, not proof. Each alert means “investigate”, never “spoofing confirmed”. Severity rises only when several signals agree, and every alert shows its evidence and possible harmless causes.

Frontend dashboard design

Illustrative mockup · placeholder data

DNScope

1 **Overview**

2 DNS Lookup

3 Response Comparison

4 Security Alerts

5 Domain History

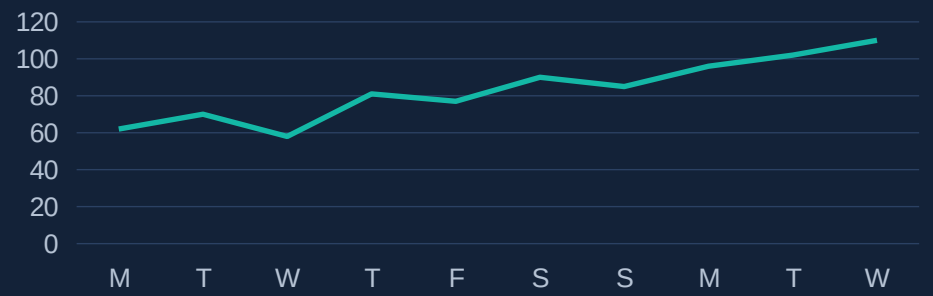
● Backend connected

Investigate a domain… e.g. shop.example.net

Check Domain

DNS checks	Domains	Open alerts	Resolver agreement
1,248	37	5	94%

DNS activity (checks / day)



Investigation: shop.example.net

MEDIUM R2 · resolver disagreement

Quad9 → 203.0.113.66 (TTL 60 s)
Others → 198.51.100.20 (TTL 300 s)
Possible: CDN routing, stale cache

Severity	Domain	Rule	Evidence	Time
MEDIUM	shop.example.net	R2	Resolvers returned different networks	10:42
HIGH	demo.example.com (sim.)	R6	Private address 10.0.0.5 in answer	09:15

Technology stack

React + TypeScript
Frontend pages and components

Tailwind CSS
Consistent dark-theme styling

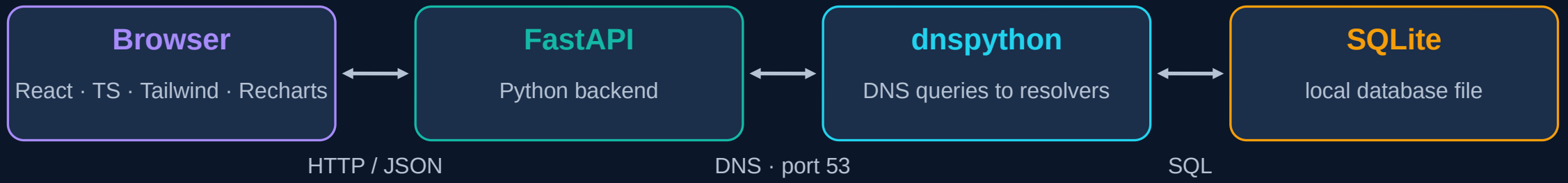
Recharts
Activity and TTL trend charts

FastAPI
Backend REST API and validation

Python + dnspython
Send DNS queries, parse records

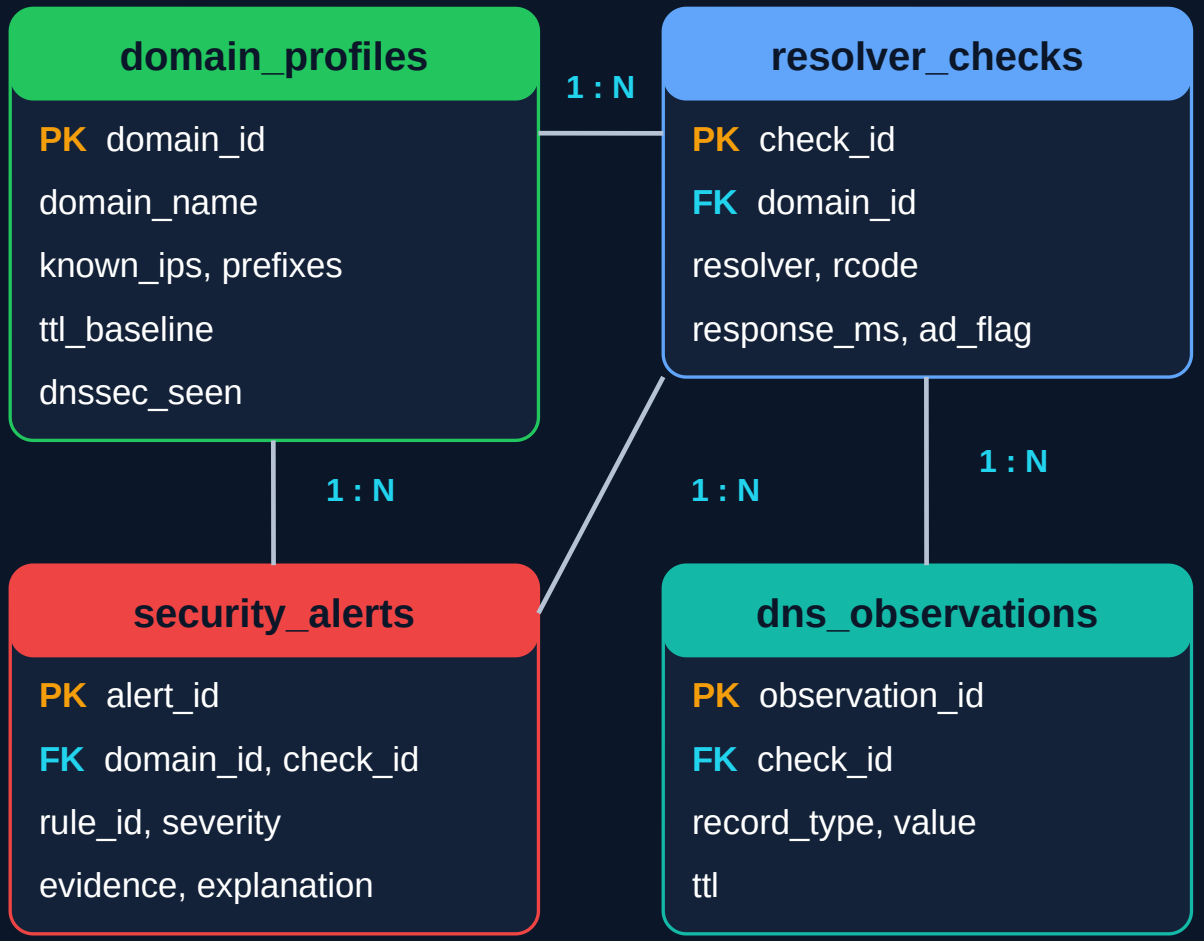
SQLite
Store observations and alerts

Git + GitHub
Version control and history



All code tracked in Git and GitHub

Database and backend



1 domain → many checks · 1 check → many records · alerts cite the observations as evidence

FastAPI endpoints

POST `/api/lookup`
DNS records for a domain

POST `/api/compare`
Query all resolvers and compare

GET `/api/alerts`
List alerts with evidence

GET `/api/domains/{domain}/history`
Past IPs, TTL trend, alerts

Only one module talks to the database; queries are parameterised.

Testing and evaluation plan

PLANNED · no results yet

Normal DNS lookup

Stable domains, repeated

→ No alert after warm-up

Historical IP change

CDN domains; simulated migration

→ No alert or Low + reason

Resolver discrepancy

Geo-DNS domains; mock resolver

→ R2 with causes listed

Invalid domain input

Empty, too long, IPs, URLs

→ Rejected with clear message

Timeout / lookup failure

Unreachable resolver, SERVFAIL

→ Partial result; R4 if repeated

DNSSEC checks

Signed and broken test zones

→ R5 only for real failures

Attack-like answers come from a labelled mock resolver, never from real attacks.

Precision

`TP / (TP + FP)`

Are the alerts warranted?

Recall

`TP / (TP + FN)`

Are real problems caught?

False-positive rate

`FP / (FP + TN)`

How often are normal domains flagged?

Detection latency

`query → alert time`

Median and 95th percentile

Targets, not results: recall ≥ 0.9 on simulated scenarios, false-positive rate < 10% on real domains, median check < 3 s — to be revised after calibration.

Differentiation, limitations and future scope

What is different

- Historical profile for each domain
- Resolver comparison combined with change detection
- Plain-language explanation for every alert
- One dashboard for the whole investigation

Limitations

- Encrypted DNS (DoH/DoT) hides traffic from passive tools; DNScope only sees its own queries
- CDN and geo-DNS changes cause false alarms
- Caching: a poisoned entry may expire before a check
- If the real server is hijacked, every resolver agrees

Future scope

- Full DNSSEC chain validation in DNScope itself
- Smarter baselines per domain and per CDN
- More test scenarios and longer measurement
- Scheduled checks for watched domains

Each technique is known; the contribution is an understandable, explainable combination — not a claim of global novelty.

Conclusion

- DNS is critical, and classic DNS answers can be forged
- DNScope compares four resolvers and each domain's history
- Transparent rules raise explained alerts, not verdicts
- Evaluation is planned with real and simulated cases

DNScope — Observe. Compare. Investigate.



Thank You

Questions?

Anwita Padhi · 24BYB0193
VIT Vellore

Key sources: RFC 1034/1035 (DNS) · RFC 3833 (DNS threats) · RFC 5452 (forged answers) · RFC 4033–4035 (DNSSEC) · RFC 7858 / 8484 (DoT / DoH)